Workshop 2: solutions

1. (a) Using the QR decomposition we have that

$$\begin{aligned}
\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|^2 &= \left\|\mathbf{y}-\mathbf{Q}\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta}\right\|^2\\
&= \left\|\mathbf{Q}^T\mathbf{y}-\mathbf{Q}^T\mathbf{Q}\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta}\right\|^2\\
&= \left\|\mathbf{Q}^T\mathbf{y}-\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta}\right\|^2\\
&= \left\|\begin{pmatrix}\mathbf{Q}_1^T\\ \mathbf{Q}_2^T\end{pmatrix}\mathbf{y}-\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta}\right\|^2\\
&= \left\|\begin{pmatrix}\mathbf{Q}_1^T\mathbf{y}-\mathbf{R}\boldsymbol{\beta}\\ \mathbf{Q}_2^T\mathbf{y}\end{pmatrix}\right\|^2\\
&= \left\|\mathbf{Q}_1^T\mathbf{y}-\mathbf{R}\boldsymbol{\beta}\right\|^2+\left\|\mathbf{Q}_2^T\mathbf{y}\right\|^2 \qquad (1)
\end{aligned}$$

Equation (1), as a function of $\boldsymbol{\beta}$, is minimised when $\mathbf{Q}_1^T\mathbf{y}=\mathbf{R}\boldsymbol{\beta}$, i.e., when $\boldsymbol{\beta}=\mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y} \Rightarrow \widehat{\boldsymbol{\beta}}=\mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y}$.

Note that $\mathbf{Q}$ in this exercise plays the role of $\mathcal{Q}$ in Section 4 of the lecture notes, whereas $\mathbf{Q}_1$ plays the role of $\mathbf{Q}$ in the notes.

(b) From the text we have that

$$\mathbf{y}\sim \mathrm{N}(\mathbf{X}\boldsymbol{\beta},\sigma^2\mathbf{I}_n).$$

From (a) we also have that

$$\begin{aligned}
\mathbb{E}(\widehat{\boldsymbol{\beta}}) &= \mathbb{E}(\mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y})\\
&= \mathbf{R}^{-1}\mathbf{Q}_1^T\mathbb{E}(\mathbf{y})\\
&= \mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{X}\boldsymbol{\beta}\\
&= \mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{Q}_1\mathbf{R}\boldsymbol{\beta}\\
&= \mathbf{R}^{-1}\mathbf{I}_p\mathbf{R}\boldsymbol{\beta}\\
&= \boldsymbol{\beta}.
\end{aligned}$$

Further, the variance-covariance matrix of $\widehat{\boldsymbol{\beta}}$ is

$$\begin{aligned}
\mathbf{V}_{\widehat{\boldsymbol{\beta}}} &= \mathrm{var}(\widehat{\boldsymbol{\beta}})\\
&= \mathrm{var}(\mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y})\\
&= \mathbf{R}^{-1}\mathbf{Q}_1^T\mathrm{var}(\mathbf{y})(\mathbf{R}^{-1}\mathbf{Q}_1^T)^T\\
&= \mathbf{R}^{-1}\mathbf{Q}_1^T(\sigma^2\mathbf{I}_n)\mathbf{Q}_1\mathbf{R}^{-T}, \qquad \mathbf{R}^{-T}=(\mathbf{R}^{-1})^T=(\mathbf{R}^T)^{-1}\\
&= \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2.
\end{aligned}$$

Because a linear combination of normal random variables follows a normal distribution, one has that

$$\widehat{\boldsymbol{\beta}} \sim \mathrm{N}(\boldsymbol{\beta}, \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2).$$

This implies that, for each $j = 1, \ldots, p$,

$$\hat{\beta}_j \sim \mathrm{N}(\beta_j, \sigma^2_{\hat{\beta}_j}), \quad \sigma^2_{\hat{\beta}_j} = [\mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2]_{jj},$$

and thus

$$\frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j}} \sim \mathrm{N}(0, 1).$$

(c) Let us start by computing the expected value of $\mathbf{Q}_2^T\mathbf{y}$

$$\begin{aligned}\mathbb{E}(\mathbf{Q}_2^T\mathbf{y}) &= \mathbf{Q}_2^T\mathbb{E}(\mathbf{y}) \\ &= \mathbf{Q}_2^T\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{Q}_2^T\mathbf{Q}_1\mathbf{R}\boldsymbol{\beta} \\ &= \mathbf{0}_{n-p},\end{aligned}$$

since $\mathbf{Q}_2^T\mathbf{Q}_1 = \mathbf{0}_{n-p,p}$ (because $\mathbf{Q}$ is orthogonal and by definition of $\mathbf{Q}_1$ and $\mathbf{Q}_2$). Regarding the covariance matrix, we have

$$\begin{aligned}\mathrm{var}(\mathbf{Q}_2^T\mathbf{y}) &= \mathbf{Q}_2^T(\sigma^2\mathbf{I}_n)\mathbf{Q}_2 \\ &= \sigma^2\mathbf{I}_{n-p}.\end{aligned}$$

Finally, and because a linear combination of normal random variables follows a normal distribution, one has that

$$\mathbf{Q}_2^T\mathbf{y} \sim \mathrm{N}(\mathbf{0}_{n-p}, \sigma^2\mathbf{I}_{n-p}).$$

(d) First note that from (1), we have that

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{Q}_1^T\mathbf{y} - \mathbf{R}\boldsymbol{\beta}\|^2 + \|\mathbf{Q}_2^T\mathbf{y}\|^2$$

and therefore

$$\begin{aligned}\|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2 &= \|\mathbf{Q}_1^T\mathbf{y} - \mathbf{R}\widehat{\boldsymbol{\beta}}\|^2 + \|\mathbf{Q}_2^T\mathbf{y}\|^2 \\ &= \|\mathbf{Q}_1^T\mathbf{y} - \mathbf{R}\mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y}\|^2 + \|\mathbf{Q}_2^T\mathbf{y}\|^2 \\ &= \|\mathbf{Q}_2^T\mathbf{y}\|^2\end{aligned}$$

From (iii) we know that each element of the vector $\mathbf{Q}_2^T\mathbf{y}$ follows a (univariate) normal distribution with mean zero and variance σ^2, that is,

$$[\mathbf{Q}_2^T\mathbf{y}]_i \sim \mathrm{N}(0, \sigma^2), \quad i = 1, \ldots, n-p,$$

or, equivalently,

$$\frac{[\mathbf{Q}_2^T\mathbf{y}]_i}{\sigma} \sim \mathrm{N}(0, 1), \quad i = 1, \ldots, n-p,$$

and thus,

$$\left(\frac{[\mathbf{Q}_2^T\mathbf{y}]_i}{\sigma}\right)^2 \sim \chi_1^2, \quad i = 1, \ldots, n-p.$$

Because the elements of the vector $\mathbf{Q}_2^T\mathbf{y}$ are independent (for the normal distribution, zero covariance implies independence), we have that

$$\begin{aligned}
\|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2 &= \frac{\|\mathbf{Q}_2^T\mathbf{y}\|^2}{\sigma^2} \\
&= \frac{\sum_{i=1}^{n-p}([\mathbf{Q}_2^T\mathbf{y}]_i)^2}{\sigma^2} \\
&= \sum_{i=1}^{n-p}\left(\frac{[\mathbf{Q}_2^T\mathbf{y}]_i}{\sigma}\right)^2 \sim \chi_{n-p}^2.
\end{aligned}$$

2. The numerator of the sample correlation is the sample covariance, which is given by

$$\frac{1}{n-1}\sum_{i=1}^{n}(\hat{\mu}_i - \bar{\hat{\boldsymbol{\mu}}})(\hat{\epsilon}_i - \bar{\hat{\boldsymbol{\epsilon}}}) = \frac{1}{n-1}\sum_{i=1}^{n}(\hat{\mu}_i\hat{\epsilon}_i - \hat{\mu}_i\bar{\hat{\boldsymbol{\epsilon}}} - \bar{\hat{\boldsymbol{\mu}}}\hat{\epsilon}_i + \bar{\hat{\boldsymbol{\mu}}}\bar{\hat{\boldsymbol{\epsilon}}}). \tag{2}$$

Note that because an intercept is included in the model we know that the residuals must sum to zero (please see Workshop 1), i.e.,

$$\sum_{i=1}^{n}\hat{\epsilon}_i = 0 \Leftrightarrow \bar{\hat{\boldsymbol{\epsilon}}} = \frac{1}{n}\sum_{i=1}^{n}\hat{\epsilon}_i = 0.$$

Therefore the last three terms in (2) vanish. Furthermore, because $\hat{\boldsymbol{\epsilon}}$ is orthogonal to $\hat{\boldsymbol{\mu}}$, we have

$$\hat{\boldsymbol{\mu}}^\mathsf{T}\hat{\boldsymbol{\epsilon}} = \sum_{i=1}^{n}\hat{\mu}_i\hat{\epsilon}_i = 0.$$

so the first term also vanishes. Hence the sample covariance is zero, and therefore the sample correlation is also zero.

3. We know that the fitted value vector when we regress $\mathbf{y}$ on $\mathbf{X}$ is $\hat{\boldsymbol{\mu}}_\mathbf{y} = \mathbf{X}\hat{\beta}_\mathbf{y} = \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y}$, where we are using the index $\mathbf{y}$ to emphasise that the response vector in this case is $\mathbf{y}$. Now, if $\hat{\boldsymbol{\epsilon}}$ is treated as the response vector and $\mathbf{X}$ as the design matrix, we have that the fitted value vector is

$$\begin{aligned}
\hat{\boldsymbol{\mu}}_{\boldsymbol{\epsilon}} &= \mathbf{X}\hat{\beta}_{\boldsymbol{\epsilon}} \\
&= (\mathbf{Q}\mathbf{R})(\mathbf{R}^{-1}\mathbf{Q}^\mathsf{T}\hat{\epsilon}) \\
&= \mathbf{Q}\mathbf{Q}^\mathsf{T}\hat{\epsilon}
\end{aligned} \tag{3}$$

But note that

$$\hat{\epsilon} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_\mathbf{y} = \mathbf{y} - \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y}$$

Returning to (3), we have that

$$\begin{aligned}
\hat{\boldsymbol{\mu}}_{\boldsymbol{\epsilon}} &= \mathbf{Q}\mathbf{Q}^\mathsf{T}(\mathbf{y} - \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y}) \\
&= \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y} - \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y} \\
&= \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y} - \mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y} \\
&= \mathbf{0}
\end{aligned}$$

This result should not come as a surprise: the residual vector is orthogonal to the column space of the design matrix, so its projection onto that space must be the zero vector.

4. The situation described in the question translates into a model of the form:

$$\mu_i = \begin{cases} kx_i + \alpha x_i^2, & \text{for alloy 1} \\ kx_i + \beta x_i^2, & \text{for alloy 2} \\ kx_i + \gamma x_i^2, & \text{for alloy 3} \end{cases}$$

where $y_i \sim N(\mu_i, \sigma^2)$. Note that we are told that when there is no load, there is no deformation; therefore the mean function does not include an intercept. Furthermore, we are told that deformation is expected to vary linearly with load in exactly the same way for all three alloys, which justifies the inclusion of the term kx_i in the mean function for each alloy. Finally, as the load increases, the three alloys deviate from linear behaviour in different ways, with the relationship becoming slightly curved. To account for this, we include three (possibly) different quadratic terms in the mean function. Let the first 6 deformation measurements correspond to the first alloy type, under the six different specified loads, the subsequent six to the second alloy type, and the final six to the third alloy type.

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ y_8 \\ y_9 \\ y_{10} \\ y_{11} \\ y_{12} \\ y_{13} \\ y_{14} \\ y_{15} \\ y_{16} \\ y_{17} \\ y_{18} \end{pmatrix} = \begin{pmatrix} x_1 & x_1^2 & 0 & 0 \\ x_2 & x_2^2 & 0 & 0 \\ x_3 & x_3^2 & 0 & 0 \\ x_4 & x_4^2 & 0 & 0 \\ x_5 & x_5^2 & 0 & 0 \\ x_6 & x_6^2 & 0 & 0 \\ x_1 & 0 & x_1^2 & 0 \\ x_2 & 0 & x_2^2 & 0 \\ x_3 & 0 & x_3^2 & 0 \\ x_4 & 0 & x_4^2 & 0 \\ x_5 & 0 & x_5^2 & 0 \\ x_6 & 0 & x_6^2 & 0 \\ x_1 & 0 & 0 & x_1^2 \\ x_2 & 0 & 0 & x_2^2 \\ x_3 & 0 & 0 & x_3^2 \\ x_4 & 0 & 0 & x_4^2 \\ x_5 & 0 & 0 & x_5^2 \\ x_6 & 0 & 0 & x_6^2 \end{pmatrix} \begin{pmatrix} k \\ \alpha \\ \beta \\ \gamma \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \\ \epsilon_7 \\ \epsilon_8 \\ \epsilon_9 \\ \epsilon_{10} \\ \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{14} \\ \epsilon_{15} \\ \epsilon_{16} \\ \epsilon_{17} \\ \epsilon_{18} \end{pmatrix}.$$

5. The null model is not *nested* within the alternative model: the predictor variable x_i appears in the null model but not in the alternative. In the notation of Section 4.3.2, the matrix $\mathbf{C}$ plays a key role in the construction of the test statistic. However, there is no such matrix $\mathbf{C}$ that consists of a selection of rows from the 4×4 identity matrix and leads to the model in H_0.

6. We can perform the required hypothesis test, whose null hypothesis is that all γ_j are zero, using an F test. Using the result from the final part of Section 4.3.2 of the notes, we obtain the F test statistic as

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/q}{\text{RSS}_1/(n-p)},$$

where RSS_1 denotes the residual sum of squares for the full model, and RSS_0 denotes the residual sum of squares for the reduced model in which H_0 holds, and q is the difference in the number of parameters between the two models. In the denominator, p denotes the number of parameters in the full model. Therefore, the denominator of the test statistic is simply the estimated variance of the error term in the full model (recall that $\hat{\sigma}^2 = \|\mathbf{r}\|^2/(n-p)$). Thus, the denominator of the F test statistic is already available to us; it is 0.3031^2. We can compute the residuals sum of squares for the full and reduced models, which are needed for the numerator, using the estimated residual standard error and the corresponding degrees of freedom. Recall that, in this context, the degrees of freedom are given by the sample size minus the number of parameters in the corresponding model. Thus, we have that $\texttt{RSS}_0 = 0.3009^2 \times 98$ and $\texttt{RSS}_1 = 0.3031^2 \times 95$. The only remaining quantity to determine is q. We do not know from the text how many γ_j's there are, but we can deduce this from the difference in degrees of freedom. The sample size is the same for both models, so the difference in the number of parameters, q, should be three. Let's now compute the value of the test statistic:

$$F = \frac{(((0.3009^2) \times 98) - ((0.3031^2) \times 95))/3}{0.3031^2} = 0.52751$$

We know that under the null hypothesis this test statistic follows a F distribution with degrees of freedom given by q and $n-p$. So, in this case the test statistics follows a $F_{3,95}$ distribution. Let's then compute the p-value.

```
pval <- pf(0.52751, 3, 95, lower.tail = F)
pval

## [1] 0.6644552
```

The p-value is 0.66 and so there is no reason to reject the null model.

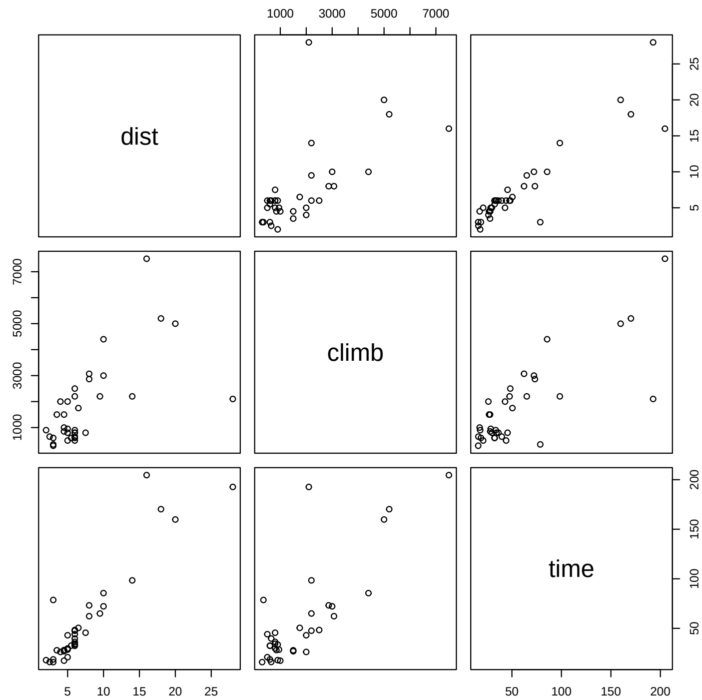
7. Statistician A is correct. The residuals versus fitted values plot suggests a mean-variance relationship (increasing spread with fitted values), violating the constant variance assumption; this directly affects standard errors, t-tests, and F-tests. The Q-Q plot assesses normality of residuals and cannot rule out heteroscedasticity. Therefore, the noncostant variance issues must be addressed regardless of how 'normal' the residuals appear.

8. The ϵ_i will almost certainly be correlated in time, violating the independence assumption of the linear model, meaning that it will not be possible to accurately assess the uncertainty of model predictions (at least, not using linear model theory). A plot of residuals against t_i should show this up or, alternatively, a plot successive pairs of residuals, say, $\hat{\epsilon}_{i+1}$ against $\hat{\epsilon}_i$, should also show the hypothesised correlation between errors (or, a bit beyond the scope of this course, one could use the R function `acf` on the residuals).

9. First, we need to load the library `MASS`. We can then begin by quickly inspecting the data. Let's do that.

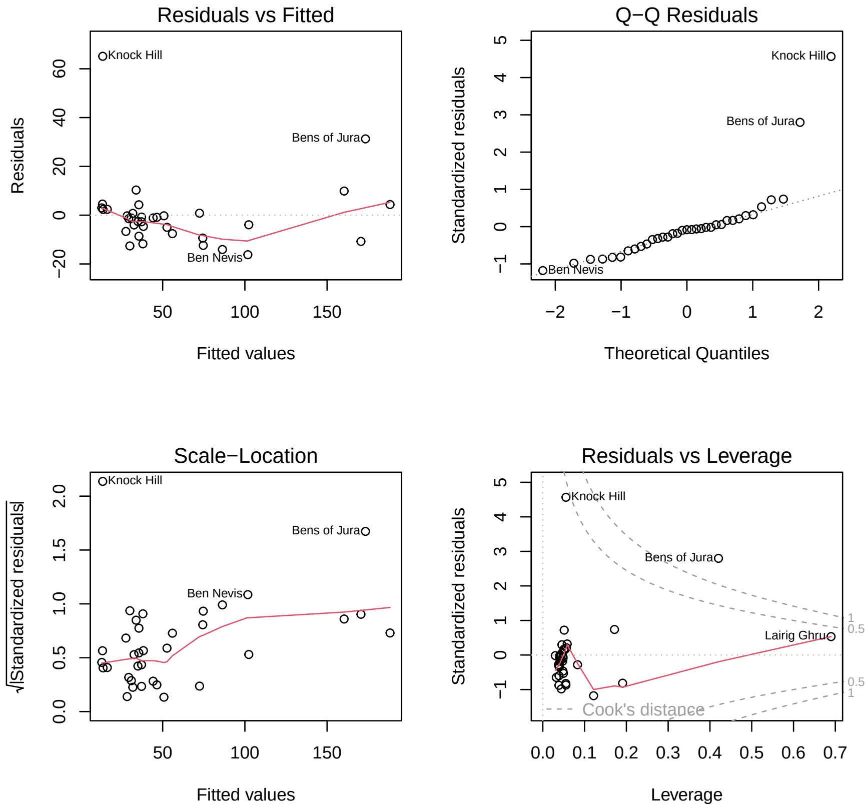
```
library(MASS)
head(hills)

##              dist climb   time
## Greenmantle   2.5   650 16.083
```

```
## Carnethy       6.0  2500 48.350
## Craig Dunain  6.0   900 33.650
## Ben Rha        7.5   800 45.600
## Ben Lomond     8.0  3070 62.267
## Goatfell       8.0  2866 73.217

plot(hills)
```



It seems natural that longer races would tend to have greater record times per mile, so we might expect the record time to be a convex increasing function of distance. However, the scatterplot of these variables reveals a quite strong linear trend. Similarly, the scatterplot of record time against climb also suggests a linear relationship. Let us now model the winning times as a function of climb (total height gained during the route) and distance, and then inspect the residual plots.

```
hm <- lm(time ~ dist + climb, data = hills)
par(mfrow = c(2, 2))
plot(hm)
```



A few remarks about the residual plots: Knock Hill has a very large standardised residual, while Bens of Jura also has a high standardised residual but remains borderline under the standard normal distribution. We can see that Bens of Jura has a Cook's distance greater than one, so it is safer to redo all analyses with and without this observation to assess how much, if anything, the results change. Let us inspect these two observations.

```
hills[which(hm$residuals > 20), ]
```

```
##              dist climb    time
## Bens of Jura   16  7500 204.617
## Knock Hill      3   350  78.650
```

Something is not right with Knock Hill; its recorded time corresponds to a slow walking speed. For this reason, we will remove this observation from the dataset, as it appears to be a recording error. We will store the new dataset in the dataframe `hills1`.

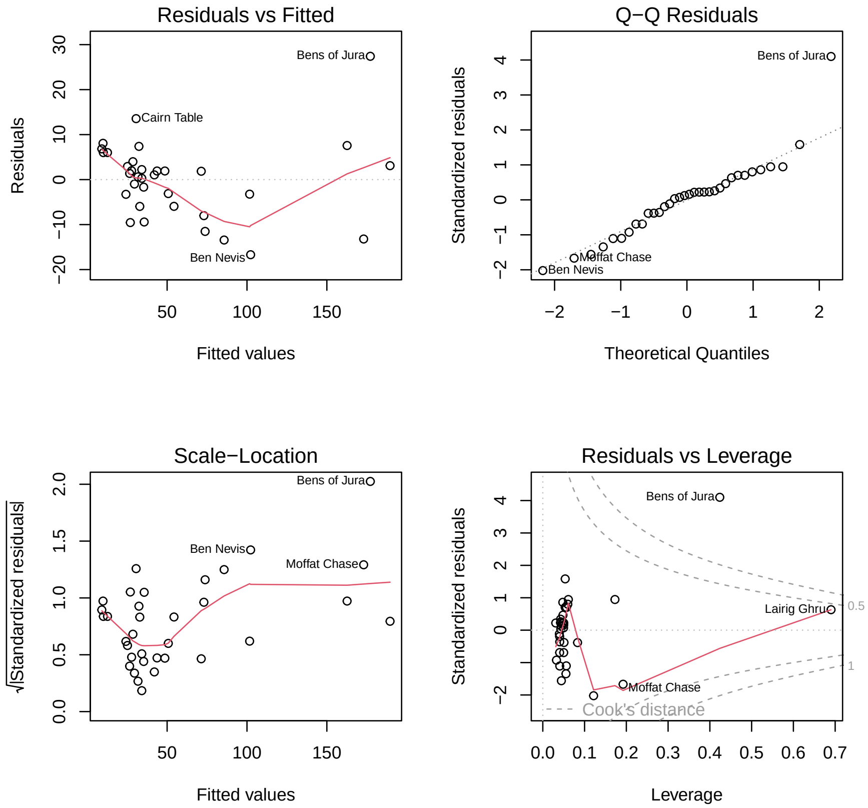
```
hills1 <- hills[- which(hm$residuals > 60), ]
```

There seems to be a trend in the residuals versus fitted values plot, although the sample size is limited (35), and for fitted values of 100 or above, there are only six observations. To better investigate, let us plot the standardised residuals against each predictor variable. We will use the version of the dataset without the big outlier (Knock Hill). Additionally, we will check the usual residual plots for this version of the dataset.

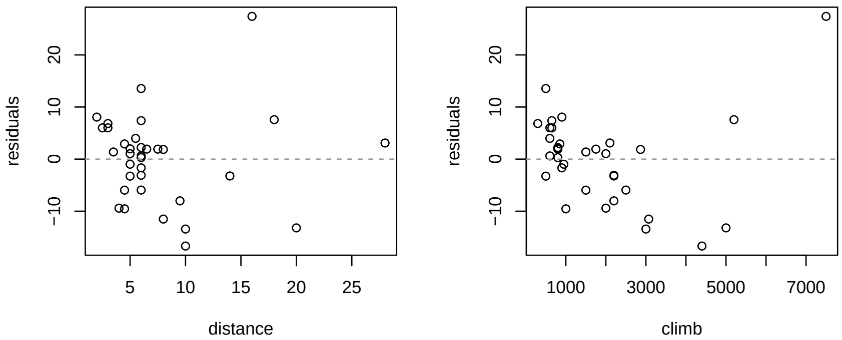
```r
hm <- lm(time ~ dist + climb, data = hills1)

summary(hm)
```

```
##
## Call:
## lm(formula = time ~ dist + climb, data = hills1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.694  -5.276   1.210   3.758  27.403
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.530352   2.649100  -5.108 1.58e-05 ***
## dist          6.364562   0.361130  17.624  < 2e-16 ***
## climb         0.011855   0.001235   9.600 8.41e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.804 on 31 degrees of freedom
## Multiple R-squared:  0.9716,Adjusted R-squared:  0.9698
## F-statistic: 530.9 on 2 and 31 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(hm)
```



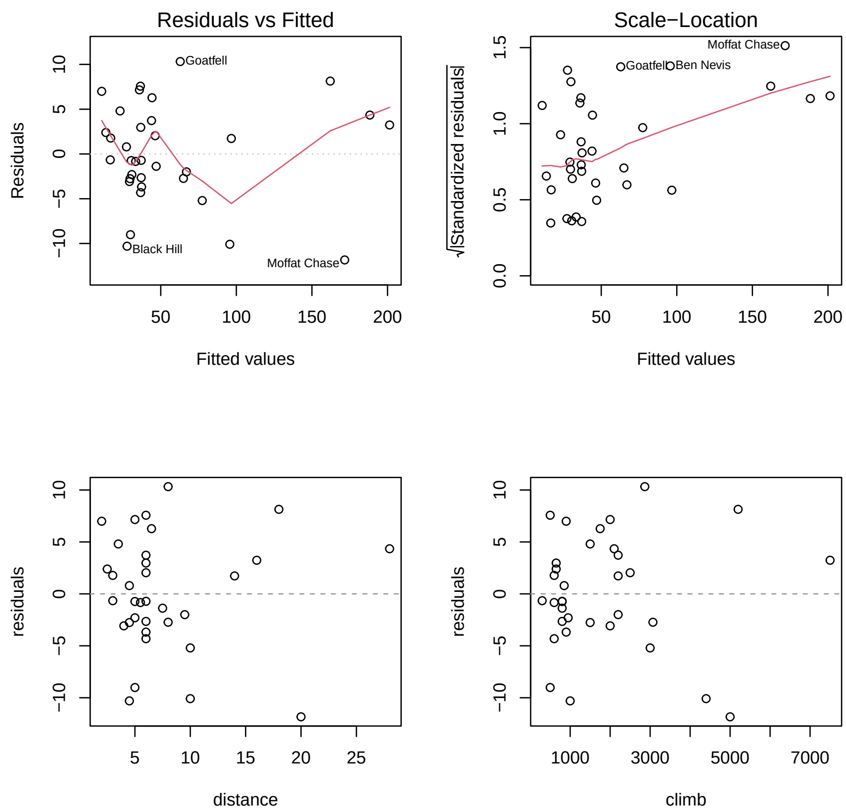

```
par(mfrow = c(2, 2))
plot(hills1$dist, residuals(hm),
     ylab = "residuals", xlab = "distance")
abline(h = 0, lty = 2, col = "gray60")

plot(hills1$climb, residuals(hm),
     ylab = "residuals", xlab = "climb")
abline(h = 0, lty = 2, col = "gray60")
```



Cook's distance's associated with Bens of Jura observation is still very large. However, it does not appear to be an erroneous observation, so we will proceed and revisit this issue later. The plot of the residuals against climb seems to indicate a slight quadratic trend, although data is sparse for climb values above 3000. Let us now include a quadratic term for this variable in the model and observe the results.

```
hm1 <- lm(time ~ dist + climb + I(climb^2), data = hills1)
summary(hm1)

##
## Call:
## lm(formula = time ~ dist + climb + I(climb^2), data = hills1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.8350  -2.7525  -0.6867   3.6026  10.3224
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.149e+00  2.354e+00  -1.338    0.191
## dist         6.597e+00  2.389e-01  27.616  < 2e-16 ***
## climb       -6.709e-04  2.085e-03  -0.322    0.750
## I(climb^2)   1.849e-06  2.838e-07   6.516 3.33e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.758 on 30 degrees of freedom
## Multiple R-squared:  0.9883,Adjusted R-squared:  0.9871
## F-statistic: 841.4 on 3 and 30 DF,  p-value: < 2.2e-16

par(mfrow = c(2, 2))
plot(hm1, which = 1)

plot(hm1, which = 3)

plot(hills1$dist, residuals(hm1),
     ylab = "residuals", xlab = "distance")
```

```r
abline(h = 0, lty = 2, col = "gray60")

plot(hills1$climb, residuals(hm1),
     ylab = "residuals", xlab = "climb")
abline(h = 0, lty = 2, col = "gray60")
```



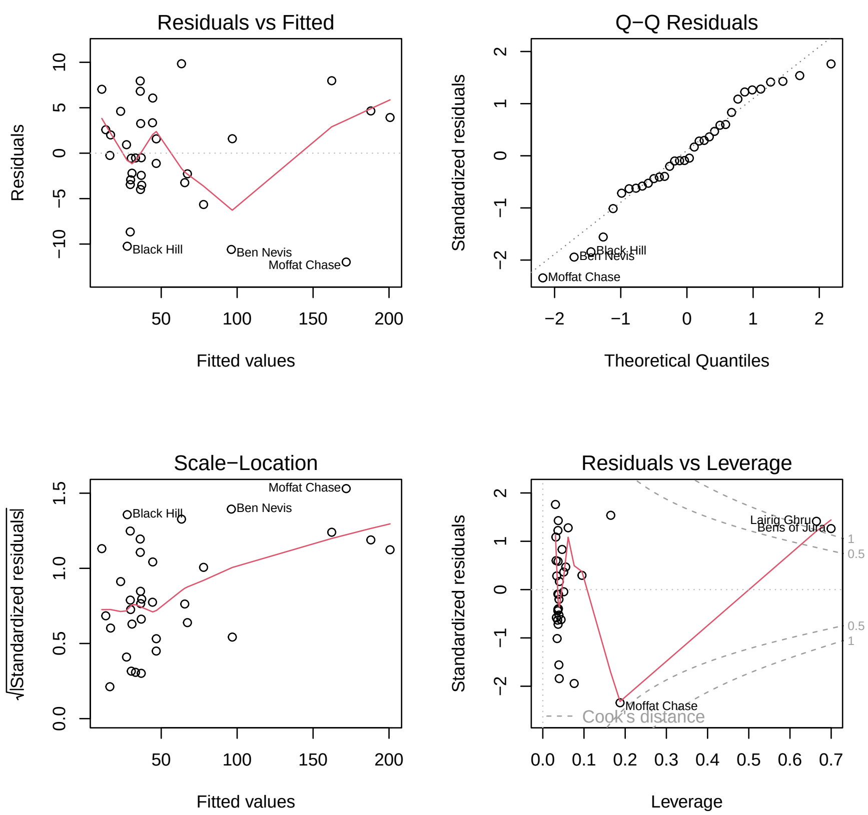

```r
AIC(hm)

## [1] 249.2575

AIC(hm1)

## [1] 221.2762
```

The residual plots appear slightly improved, and the adjusted R squared has also increased (and AIC has decreased). However, the effect of climb is no longer significant. Let us drop it and check the results.

```r
hm2 <- lm(time ~ dist + I(climb^2), data = hills1)
summary(hm2)

##
```

```
## Call:
## lm(formula = time ~ dist + I(climb^2), data = hills1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.9794  -3.1662  -0.3776   3.7822   9.8419
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.663e+00  1.703e+00  -2.151   0.0393 *
## dist         6.568e+00  2.173e-01  30.225   <2e-16 ***
## I(climb^2)   1.765e-06  1.083e-07  16.293   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.674 on 31 degrees of freedom
## Multiple R-squared:  0.9882,Adjusted R-squared:  0.9875
## F-statistic:  1300 on 2 and 31 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(hm2)
```



```
AIC(hm2)

## [1] 219.3934
```

All terms are now significant, and the adjusted R squared is comparable (even slightly higher). AIC is also slightly lower for this model. Based on this model, let us now predict the winning time for a 7 mile race with 2400 feet of ascent.

```
coef(hm2)[1] + 7*coef(hm2)[2] + (2400^2)*coef(hm2)[3]

## (Intercept)
##    52.47657
```

Using what we will cover next week, we can also compute a prediction interval.

```
predict(hm2, newdata = data.frame(dist = 7, climb = 2400),
        interval = "prediction")

##        fit      lwr      upr
## 1 52.47657 40.73149 64.22165
```

Let us see how this result changes when the two observations with a Cook's distance greater than one are removed from the dataset. These observations are number 7 (Bens of Jura) and number 11 (Lairig Ghru).

```
hills2 <- hills1[-c(7, 11), ]
hm3 <- lm(time ~ dist + I(climb^2), data = hills2)
predict(hm3, newdata = data.frame(dist = 7, climb = 2400),
        interval = "prediction")

##        fit      lwr      upr
## 1 52.13239 40.40349 63.86128
```

As can be observed, the results are essentially the same. A possible short report is as follows.

Data on winning times for 34 Scottish hill races were analysed by linear modelling (a 35th time was excluded from the original dataset as it appears to represent a recording error). In particular the following simple model was used:

$$\texttt{time}_i = \beta_0 + \beta_1 \texttt{dist}_i + \beta_2 \texttt{climb}_i^2 + \epsilon_i \tag{4}$$

where `time` was winning time in minutes, `dist` was overall distance of race and `climb` was total ascent in the race. ϵ_i represents random variability in the times unexplained by the distance and height variables. The β_j are parameters of the model, which were estimated by fitting the model to the 34 data, by least squares. The data set analysed contained races with a range of distances and heights covering those proposed for the Whisky challenge, suggesting that the model should be reasonably reliable for predictive purposes.

The model structure was arrived at by starting with a model with linear dependence on `dist` and `climb`, but this was (slightly) improved by the model structure in (4). This model would predict a winning time of 52.5 minutes for the Whisky Challenge, with the 95% prediction interval estimate for the wining time being between 40.7and 64.2 minutes. However you should be aware that the act of offering substantial prizes for achieving particular times will in itself make the Whisky Challenge somewhat different to the other races used to estimate the model.

Note: This may not be the only possible correct solution.